# UrbanFlow AI: An Uncertainty-Aware and Explainable Spatio-Temporal Graph Neural Network Framework for Robust Traffic Forecasting

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***Abstract*—Urban traffic forecasting is an important component of intelligent transportation systems because accurate prediction of future traffic conditions can support congestion management, travel-time estimation, and proactive traffic control. Recent advances in Spatio-Temporal Graph Neural Networks (STGNNs) have significantly improved traffic forecasting by jointly modelling spatial dependencies between road sensors and temporal patterns in traffic observations. However, recent literature shows that existing approaches increasingly focus on dynamic graph learning, long-term temporal modelling, attention mechanisms, and Transformer architectures, while challenges related to uncertainty, distribution shifts, and interpretability remain important.**

**This paper reports the development and experimental study of UrbanFlow AI, a traffic forecasting research pipeline based on Spatio-Temporal Graph Neural Networks (STGNNs). Experiments use the METR-LA benchmark, with 207 sensors sampled at five-minute intervals, a 12-step historical input, and a three-step forecasting horizon. The study evaluates conventional forecasting baselines, an STGCN baseline, a multi-scale temporal STGCN, temporal-kernel ablations, timestamp-derived context variants, and Monte Carlo Dropout (MC-Dropout) uncertainty estimates. A validation-based scalar calibration procedure is used to adjust predictive intervals. Post-hoc sensitivity analyses use temporal and sensor occlusion to measure changes in model predictions, and a separate synthetic perturbation experiment examines sensitivity to imposed speed reductions. Results show that model complexity does not guarantee improvement across all point-forecasting metrics, and that uncalibrated MC-Dropout intervals can be substantially under-dispersed. Calibration improves empirical coverage at the cost of wider intervals. Occlusion experiments indicate greater sensitivity to recent observations and heterogeneous sensitivity across sensors. These findings are empirical results from a single benchmark; they do not establish causal explanations or generalization to other regions.**

***Index Terms*—traffic forecasting, graph neural networks, spatio-temporal graph neural networks, uncertainty estimation, explainable AI, distribution shift, intelligent transportation systems**

## I. Introduction

Urban transportation networks generate large volumes of spatially distributed and temporally evolving data. Traffic conditions at one road sensor are influenced not only by its recent history but also by conditions at neighbouring and upstream locations. Traffic forecasting is therefore fundamentally a spatio-temporal learning problem.

Traditional traffic forecasting methods have used statistical models, time-series approaches, and conventional machine learning techniques. However, these approaches can struggle to represent the complex spatial dependencies present in road networks. The development of Graph Neural Networks (GNNs) has therefore provided a natural framework for representing transportation networks as graphs in which sensors or intersections form nodes and roads or relationships form edges.

Spatio-Temporal Graph Convolutional Networks (STGCNs) demonstrated that graph convolution and temporal convolution can be combined to capture both spatial and temporal traffic dependencies [1]. Diffusion Convolutional Recurrent Neural Networks (DCRNN) subsequently modelled traffic propagation using diffusion convolution and recurrent temporal dynamics [2]. Other approaches have investigated adaptive graph learning, attention mechanisms, dynamic graph construction, and Transformer architectures [3]–[6].

Recent research has increasingly moved beyond static spatial graphs. Dynamic graph models attempt to represent changing relationships between traffic locations, while Transformer-based architectures attempt to capture long-range temporal dependencies. Models such as DST-GTN, STGormer, LSTTN, and HSTGNN demonstrate the increasing sophistication of modern traffic forecasting systems [11]–[13], [19].

Despite these advances, predictive accuracy alone does not necessarily make a forecasting system suitable for practical transportation applications. A traffic management system should ideally indicate how reliable a prediction is, particularly when the current traffic conditions differ from those represented in the training data. Distribution shifts can occur because of changes in traffic demand, road usage, incidents, weather, events, or other external conditions.

Recent research has begun explicitly investigating robustness to distribution shifts. For example, recent work has demonstrated that traffic forecasting models can experience significant performance degradation under spatial or temporal distribution changes [21], [22]. At the same time, explainable

traffic forecasting approaches have begun incorporating self-explanation and counterfactual reasoning [23], [24].

Uncertainty estimation, robustness, and explanation are nevertheless often treated as separate problems. This motivates the investigation of a unified forecasting framework capable of producing a prediction together with an indication of uncertainty and an interpretable description of the spatial-temporal factors influencing that prediction.

This research investigates an uncertainty-aware and explainable STGNN framework for traffic forecasting. The study records baseline performance on a standard traffic forecasting benchmark and evaluates an STGNN architecture through several experiment families. Because some notebook generations use different chronological split sizes, not every recorded model score can be treated as a matched controlled comparison. The paper distinguishes within-run ablations from historical cross-run summaries.

The objectives of this work are:

- To establish reproducible baseline traffic forecasting models.
- To implement an established STGNN architecture such as STGCN.
- To investigate uncertainty estimation for traffic predictions.
- To investigate spatially grounded explanations for predictions.
- To evaluate robustness under distribution-shift conditions.
- To conduct ablation experiments to isolate the contribution of each proposed component.

## II. Literature Review

### A. Early Spatio-Temporal Graph Neural Networks

The application of graph neural networks to traffic forecasting was established through several influential architectures. STGCN introduced a convolutional architecture that combines graph convolution with temporal convolution to model spatial and temporal dependencies simultaneously [1]. The model established a relatively simple and computationally efficient foundation for subsequent STGNN research.

DCRNN introduced diffusion convolution to represent the directional propagation of traffic information through road networks and combined this mechanism with recurrent neural networks [2]. This approach highlighted the importance of modelling traffic as a diffusion process rather than relying only on conventional spatial convolution.

Graph WaveNet subsequently introduced adaptive dependency learning, allowing the model to learn relationships between nodes in addition to using predefined graph structure [3]. This development was important because physical road connectivity does not necessarily represent all statistically useful traffic dependencies.

ASTGCN introduced spatial and temporal attention mechanisms to enable the model to dynamically focus on relevant locations and time periods [4]. Attention-based approaches consequently became an important direction in traffic forecasting.

### B. Adaptive and Dynamic Graph Learning

A major research direction has been the replacement of fixed graph structures with learned or adaptive relationships. AGCRN introduced node-specific adaptive graph learning and combined it with recurrent temporal modelling [5]. MTGNN similarly demonstrated that graph structure can be learned directly from multivariate time-series data rather than being completely specified beforehand [6].

Subsequent models have moved from static adaptive graphs toward explicitly dynamic graph construction. DGCRN investigates dynamic graph convolution for traffic prediction, while DSTAGNN and D2STGNN further explore dynamic spatial-temporal relationships [7]–[9].

Recent work continues this direction. DST-GTN introduces dynamic spatio-temporal features and dynamic adjacency relationships within a graph Transformer architecture [11]. MFGCN learns node-specific temporal relationships and time-specific spatial relationships [15]. HSTGNN combines dynamic graph generation with temporal frequency decomposition and heterogeneous temporal modelling [19].

These developments indicate that dynamic spatial modelling is now a well-established research direction. Therefore, simply replacing a static graph with a dynamic graph would provide limited novelty for a new research contribution.

### C. Long-Term and Multi-Scale Temporal Modelling

Another major limitation investigated in recent work is the difficulty of modelling long-term traffic dependencies. Short historical windows may fail to capture periodic patterns and long-term trends.

LSTTN addresses this issue through separate long- and short-term temporal modelling components and evaluates long-horizon traffic forecasting [13]. Other approaches use multi-scale temporal convolutions, frequency-domain decomposition, and hierarchical representations.

DSTSPYN uses a spatial-temporal similarity pyramid and multi-scale temporal modelling to improve the extraction of long-term dependencies. HSTGNN also introduces frequency-domain temporal decoupling and dual-channel temporal learning [16], [19].

These approaches demonstrate that long-term and multi-scale temporal modelling are already actively investigated. Consequently, a research contribution based solely on adding a long-term temporal module would also have limited novelty.

### D. Transformer-Based Traffic Forecasting

Transformers have become increasingly common in traffic forecasting because of their ability to model long-range dependencies.

STGormer uses a graph Transformer framework to address spatio-temporal heterogeneity and introduces a mixture-of-experts mechanism to handle different traffic patterns [12]. STFEformer combines spatial and temporal attention with fusion embeddings and graph-masked attention [18]. DST-GTN also uses a graph Transformer together with dynamic spatial-temporal relationships [11].

Recent work has additionally considered the computational cost of Transformer-based spatial modelling. PatchSTG proposes spatial patching to improve the scalability of large-scale traffic Transformers [20].

The literature therefore suggests that Transformer-based spatial and temporal attention is a mature direction. A new contribution should instead identify a specific limitation of current Transformer/STGNN systems.

### *E. Robustness and Distribution Shift*

Most benchmark traffic forecasting studies evaluate models using training and testing data drawn from similar distributions. In real transportation systems, however, traffic patterns can change across time and locations.

Recent research has explicitly investigated spatial distribution shifts. The work "Robust Traffic Forecasting against Spatial Shift over Years" shows that standard traffic forecasting models can experience significant degradation when evaluated under cross-year spatial shifts [21].

Another recent approach, Robust Traffic Forecasting with Disentangled Spatiotemporal Graph Neural Networks, introduces invariant representations intended to improve forecasting under distribution shifts [22].

These studies establish that robustness is a real limitation of existing STGNNs. At the same time, they show that generic "robust traffic forecasting" is already becoming a research area. Therefore, the present work must define a narrower and measurable robustness setting.

### *F. Explainability in Traffic Forecasting*

Interpretability is another important consideration for transportation applications. SEGCRN integrates explanation mechanisms directly into the forecasting architecture rather than applying a separate post-hoc explanation stage [23].

Counterfactual explanation approaches have also been proposed for traffic forecasting. Such methods attempt to determine what input conditions would need to change in order to alter a prediction [24].

These approaches demonstrate that explainability is feasible for traffic forecasting. However, there remains an opportunity to investigate whether explanations can be made spatially grounded, uncertainty-aware, and useful under distribution shifts.

## III. Research Gap and Motivation

The literature review reveals several important observations.

First, dynamic graph learning has become a well-established component of modern traffic forecasting systems. Second, long-term and multi-scale temporal modelling has received substantial attention. Third, Transformer-based attention mechanisms are increasingly common. Consequently, these mechanisms individually do not provide a sufficiently strong basis for a new contribution.

A second observation concerns robustness. Recent studies demonstrate that traffic forecasting models can degrade when the distribution of traffic observations changes. Such changes may arise across different years, locations, traffic regimes, or unusual conditions. However, standard benchmark evaluation generally emphasizes average prediction accuracy rather than prediction reliability under distribution changes.

A third observation concerns explainability. Existing work has explored self-explainable models and counterfactual explanations, but explainability is frequently evaluated separately from uncertainty and robustness.

Based on these observations, this research investigates the following research gap:

> *Existing STGNN traffic forecasting methods provide increasingly accurate predictions and have separately explored dynamic graph learning, robust forecasting, and explainability; however, a unified framework that provides traffic predictions together with uncertainty estimates and spatially grounded explanations under distribution-shift conditions remains comparatively underexplored.*

This gap will be validated experimentally rather than assumed to be solved by architectural complexity alone. The study is organized around six research questions:

- **RQ1:** How do the recorded forecasting results compare with conventional baselines and an STGCN reference?
- **RQ2:** Does parallel multi-scale temporal modelling improve forecasting relative to single-kernel alternatives?
- **RQ3:** Do timestamp-derived context features improve prediction, and does the integration mechanism matter?
- **RQ4:** How strongly are MC-Dropout uncertainty estimates associated with absolute forecasting error?
- **RQ5:** Does validation-based scalar calibration bring empirical test coverage closer to the nominal 95% level, and at what interval-width cost?
- **RQ6:** How sensitive are predictions to occlusion of historical time steps and sensor inputs?

The answers are deliberately scoped to the experiment families and splits actually recorded in the notebooks. In particular, RQ1 is interpreted cautiously because baseline notebook generations do not all document the same split sizes.

## IV. Problem Formulation

The completed experiments instantiate the general formulation with $N = 207$ nodes and $F = 1$ traffic feature per node. The historical window is $T = 12$ observations and the forecast horizon is $H = 3$. Consequently, each sample is represented as

$$X \in \mathbb{R}^{12\times 207\times 1} \tag{1}$$

and the corresponding target is

$$Y \in \mathbb{R}^{3\times 207\times 1}. \tag{2}$$

For a batch of size $B$, the model receives $(B, 12, 207, 1)$ and produces $(B, 3, 207, 1)$. Maintaining these dimensions throughout preprocessing, training, evaluation, and inference is an explicit implementation invariant.

TABLE I

QUALITATIVE COMPARISON OF REPRESENTATIVE TRAFFIC-FORECASTING APPROACHES. ENTRIES SUMMARIZE CAPABILITIES DESCRIBED IN THE CITED MODEL PAPERS; "NOT A STATED FOCUS" DOES NOT IMPLY THAT A METHOD CANNOT BE EXTENDED WITH THAT CAPABILITY.

| Method | Spatial modelling | Temporal modelling | Adaptive graph learning | Predictive uncertainty | Explainability |
|---|---|---|---|---|---|
| STGCN [1] | Graph convolution on supplied graph | Temporal convolution | No learned adaptive graph in the standard formulation | Not a stated focus | Not a stated focus |
| DCRNN [2] | Directional diffusion convolution | Recurrent encoder–decoder | Uses supplied directed graph | Not a stated focus | Not a stated focus |
| Graph WaveNet [3] | Fixed graph and learned node dependencies | Dilated causal temporal convolution | Yes, adaptive dependency matrix | Not a stated focus | Not a stated focus |
| AGCRN [5] | Node-adaptive graph convolution | Recurrent temporal modelling | Yes, node embeddings induce adaptive dependencies | Not a stated focus | Not a stated focus |
| UrbanFlow AI (this study) | Fixed METR-LA row-normalized adjacency | Parallel temporal kernels (3, 5) in two STGCN-style blocks | No | MC-Dropout with validation-selected scalar interval calibration | Post-hoc temporal and sensor occlusion sensitivity |

For a normalized traffic matrix $Z$ and starting index $s$, the chronological sliding-window construction is

$$X_s = Z_{s:s+12}, \qquad Y_s = Z_{s+12:s+15}. \tag{3}$$

Thus, the last historical observation immediately precedes the first target observation. No target observation is included in the input window.

The final training, validation, and test partitions contain 23,976, 3,413, and 6,841 sequences, respectively. The partitions remain chronological throughout the experiments.

Per-sensor standardization is fitted only on the training observations:

$$z_{i,t} = \frac{x_{i,t} - \mu_i}{\sigma_i}. \tag{4}$$

The same training-derived $\mu_i$ and $\sigma_i$ are applied to validation and test observations. Reported traffic-scale metrics are calculated after inverse transformation.

A traffic network is represented as a graph

$$G = (V, E, A) \tag{5}$$

where $V$ represents traffic sensors or road intersections, $E$ represents spatial relationships, and $A$ is the corresponding adjacency matrix.

At each time step $t$, the traffic observation is represented by

$$X_t \in \mathbb{R}^{N \times F} \tag{6}$$

where $N$ is the number of traffic nodes and $F$ is the number of features.

Given a historical sequence of observations

$$X_{t-T+1}, \ldots, X_t \tag{7}$$

the forecasting objective is to predict future traffic states

$$\hat{Y}_{t+1}, \ldots, \hat{Y}_{t+H}. \tag{8}$$

The forecasting backbone produces the point prediction $\hat{Y} = f_\theta(X, G)$. Uncertainty is estimated in a separate post-training MC-Dropout evaluation by repeated stochastic forward passes. The explanation analysis is also post-hoc: temporal and sensor inputs are occluded and the resulting prediction changes are measured. Thus, neither uncertainty nor occlusion explanations are additional outputs of the deterministic forecasting function $f_\theta$ itself.

## V. PROPOSED METHODOLOGY

### A. Overview

The completed implementation can be viewed as a sequence of deterministic data transformations followed by learned forecasting and uncertainty estimation. The data pipeline validates the raw traffic matrix and timestamps, constructs the graph in the same sensor ordering, performs a chronological train/validation/test split, fits training-only normalization statistics, and generates fixed-length forecasting sequences.

The model stage applies temporal feature extraction and graph-based spatial aggregation. The proposed architecture extends the established STGCN design by using parallel temporal kernels of sizes 3 and 5. Their outputs are concatenated before graph propagation. Two such blocks are followed by a node-wise forecasting head that maps the historical representation to three future steps. A completed contextual ablation additionally investigates temporal context integration using hour-of-day, day-of-week, and weekend features. Two integration strategies are compared: direct input concatenation and context-conditioned gating.

The uncertainty stage is deliberately separated from point forecasting. The trained model is evaluated repeatedly with

dropout active, producing stochastic forecasts. Their mean provides the predictive estimate, while their standard deviation provides the raw uncertainty estimate. Empirical coverage is evaluated before and after calibration.

The proposed UrbanFlow framework extends an established STGNN baseline with uncertainty estimation and spatially grounded explanation components.

The overall architecture is:

### B. Spatio-Temporal Encoder

The completed STGCN baseline uses two spatio-temporal blocks. Each block contains temporal convolution, graph convolution, a second temporal convolution, nonlinear activation, and dropout. Temporal operations are applied along the historical dimension, while graph convolution mixes information between sensor nodes.

The supplied METR-LA graph contains 207 nodes and 1,722 non-zero entries, with density approximately 0.0402. Traffic sensor identifiers were checked against the identifiers in the graph file so that tensor and adjacency ordering remain consistent.

The graph is row-normalized directly:

$$A_{ij}^{norm} = \frac{A_{ij}}{\sum_j A_{ij}}. \tag{9}$$

All nodes have positive degree and normalized row sums equal one within numerical tolerance. The supplied adjacency is not symmetrized, so its original directional weighting is retained.

For an intermediate representation $H$, graph propagation is implemented as

$$H'_{b,t,i,:} = \sum_j A_{ij}^{norm} H_{b,t,j,:} W, \tag{10}$$

where $W$ is a learned linear projection. The STGCN baseline uses 32 hidden channels and kernel size 3. Its forecasting head maps the historical hidden representation to the three forecast steps.

The implemented STGCN serves as the primary established graph-based baseline. Temporal convolution extracts local temporal patterns, while graph convolution propagates information between connected traffic nodes. The reported implementation uses two spatio-temporal blocks, 32 hidden channels, temporal kernel size 3, dropout probability 0.2, and a 12-step-to-3-step forecasting head.

For clarity, the expression above is the operation used in this implementation; it should not be replaced by the symmetric GCN normalization $D^{-1/2}AD^{-1/2}$. In particular, the row-normalized operator is generally not symmetric, and the two normalizations encode different propagation rules. The implementation-level equation is therefore retained throughout the paper:

$$H_{b,t,:,:}^{(l+1)} = \sigma\Big(A^{\mathrm{norm}} H_{b,t,:,:}^{(l)} W^{(l)}\Big), \qquad A^{\mathrm{norm}} = D^{-1}A. \tag{11}$$

### C. Context-Aware Forecasting Ablation

The completed contextual experiment investigates whether temporal context provides additional predictive information beyond historical traffic and the fixed spatial graph. Five deterministic temporal features are constructed from the timestamps: hour-of-day sine and cosine, day-of-week sine and cosine, and a weekend indicator. These features are aligned only with the historical 12-step input window, so no future traffic observation is used as an input.

Two context-integration strategies are evaluated against the same traffic-only Multi-Scale STGCN. The first concatenates the five context features with the traffic feature at the model input. The second uses a learned context gate after the first spatio-temporal block. The gate maps the five temporal features to the hidden dimension and multiplicatively modulates the learned traffic representation. Both variants retain the same graph, forecasting horizon, training protocol, and evaluation procedure.

The context-aware experiment is therefore treated as a controlled architectural ablation rather than as a replacement for the primary Multi-Scale STGCN result. This distinction is important because the direct concatenation strategy did not improve the held-out test metrics, whereas the gated strategy reduced RMSE while producing slightly higher MAE and MAPE.

### D. Uncertainty Estimation

The completed uncertainty component uses Monte Carlo Dropout. Thirty stochastic forward passes are generated while the trained dropout layers remain active. The resulting predictive samples are summarized by their mean and standard deviation. Nominal 95% intervals are first evaluated without calibration and are then recalibrated using a scalar factor selected exclusively on the validation set. This separates uncertainty estimation from uncertainty calibration and avoids using test targets to select the calibration parameter.

### E. Post-hoc Temporal and Sensor Sensitivity

The current explainability experiment uses occlusion as a post-hoc sensitivity analysis. For temporal occlusion, one historical input step is replaced at a time and the resulting change in the model prediction is measured. For sensor occlusion, one sensor's input series is occluded and the absolute change in predictions is aggregated. These scores indicate model sensitivity to the perturbation; they are not causal effects and do not prove that a sensor controls traffic conditions.

The temporal analysis was run on up to 512 test samples. Mean absolute prediction change increased toward the most recent input steps. The largest mean change occurred at the latest step (8.8339), followed by the previous step (5.7420) and the step two positions before the forecast (3.1121). The corresponding 95th-percentile changes were 45.9813, 31.0746, and 13.7981. The pattern indicates that the trained model is more sensitive, on average, to recent observations than to distant observations in the 60-minute input window.

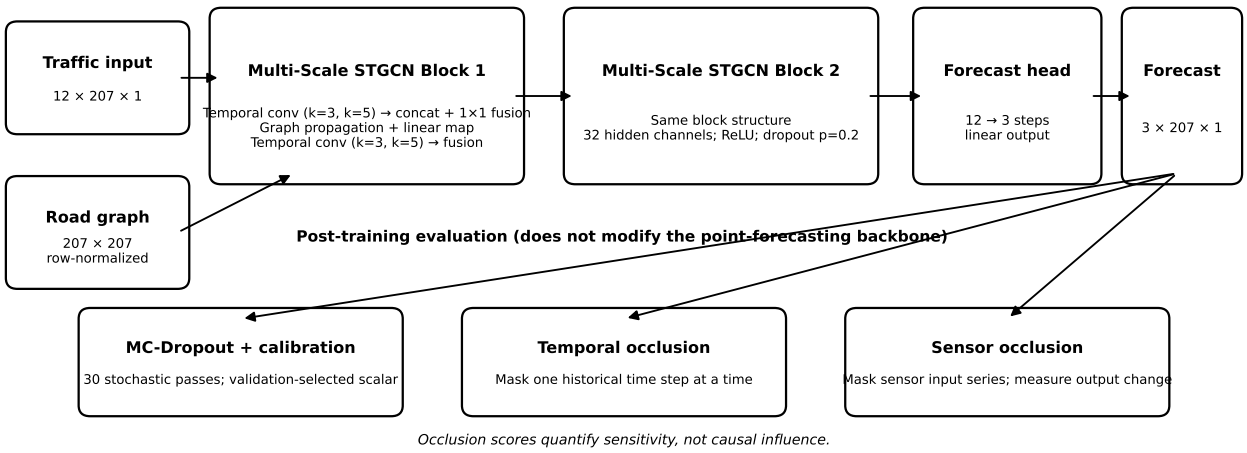



Fig. 1. Implementation-grounded UrbanFlow architecture. The forecasting backbone contains two Multi-Scale STGCN blocks. Each block applies parallel temporal convolutions with kernel sizes 3 and 5, feature concatenation and $1 \times 1$ fusion, graph propagation using the row-normalized sensor adjacency, and a second multi-scale temporal transformation. The forecasting head maps 12 historical steps to three future steps. MC-Dropout and occlusion analyses are post-training evaluation branches.

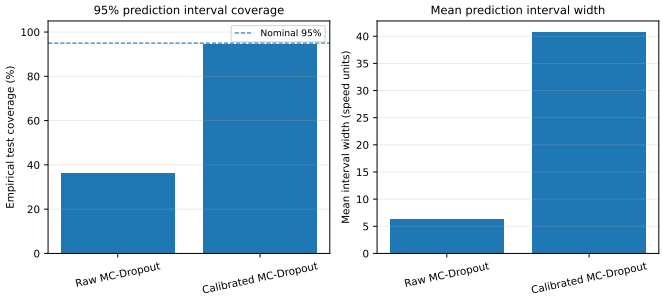



Fig. 2. Raw and validation-calibrated MC-Dropout interval performance on the held-out test set. Calibration moves empirical coverage toward the nominal 95% target but substantially increases mean interval width.

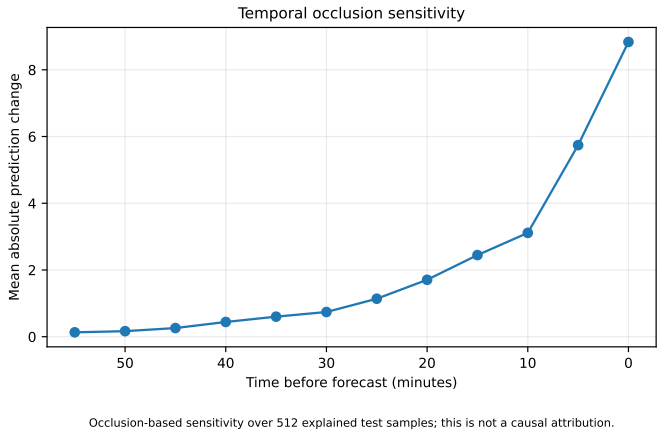



Fig. 3. Temporal occlusion sensitivity. Masking more recent historical observations produces larger mean absolute prediction changes in the evaluated samples.

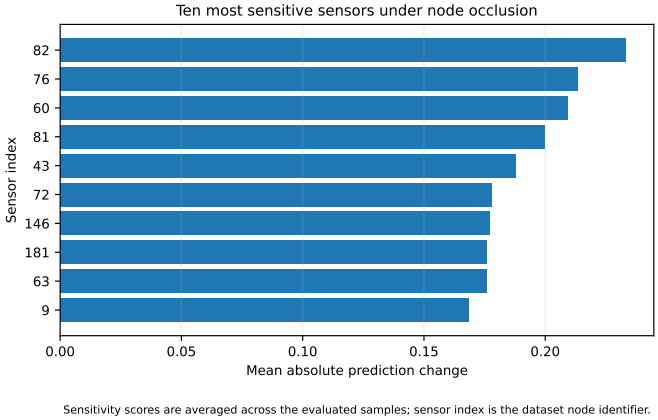



Fig. 4. Top ten sensor nodes ranked by mean absolute prediction change under sensor occlusion. These scores indicate model sensitivity, not causal influence.

TABLE II
MOST SENSITIVE SENSORS IN THE COMPLETED OCCLUSION ANALYSIS.

| Sensor index | Sensor ID | Latitude | Longitude |
|---|---|---|---|
| 82 | 769430 | 34.15818 | -118.45658 |
| 76 | 717499 | 34.15666 | -118.44808 |
| 60 | 769444 | 34.15555 | -118.43908 |
| 81 | 769431 | 34.15843 | -118.45664 |
| 43 | 760024 | 34.15930 | -118.46483 |

Sensor indices were mapped to sensor identifiers using the ordering in the dataset's adjacency metadata. Coordinates were joined by sensor ID against a METR-LA sensor-location table, rather than assuming that row order was consistent between files. The five sensors above were among the highest-scoring sensors in the occlusion run. Their coordinates enable geographic inspection, but the current experiment does not establish that these locations are the most congested, operationally important, or causally influential parts of the network.

The temporal and sensor occlusion results provide an initial explanation of which input portions affect the model's outputs. However, occlusion scores depend on the chosen replacement value, evaluated samples, output aggregation, and model checkpoint. Future work should test explanation stability across random seeds and checkpoints, compare multiple perturbation baselines, and evaluate whether the identified sensors are useful to traffic operators.

### *F. Distribution-Shift Evaluation*

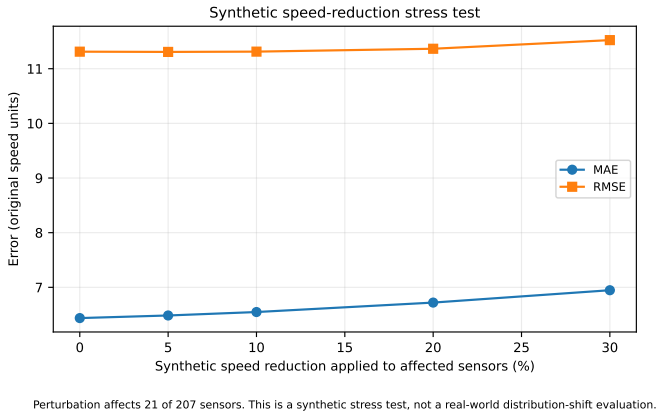



Fig. 5. Synthetic speed-reduction stress test. Errors increase with larger perturbations, although the experiment is synthetic and should not be interpreted as evidence of real-world distribution-shift robustness.

A controlled synthetic stress test was implemented by applying speed-reduction perturbations to a subset of the traffic inputs and evaluating the resulting predictions. This experiment is a limited sensitivity test, not a benchmark of real incidents or a complete distribution-shift evaluation. The perturbation procedure and the exact checkpoint must be recorded alongside the results because a synthetic speed reduction does not reproduce all characteristics of real congestion, sensor faults, or external disruptions.

In one reconstructed stress-test run, the reference metrics were MAE 6.4373, RMSE 11.3119, and MAPE 14.0925%. Under perturbation levels of 5%, 10%, 20%, and 30%, MAE increased by approximately 0.74%, 1.72%, 4.41%, and 7.92%, respectively. At 30% perturbation, RMSE increased by approximately 1.87% and MAPE by 3.47%. These figures belong to that specific reconstructed run and should not be merged with the primary model table unless checkpoint, preprocessing, and metric-scale consistency are verified.

Potential extensions include:

- temporal distribution shifts,
- high-congestion periods,
- sudden traffic changes,
- missing observations,
- noisy observations.

The exact shift protocol will depend on the selected benchmark dataset.

## VI. Proposed Algorithm

The proposed algorithm can be summarized as follows.

**Algorithm 1** UrbanFlow Uncertainty-Aware Explainable STGNN

Input historical traffic sequence $X$
Load road graph $G = (V, E, A)$
Normalize traffic observations
Generate spatio-temporal representation
Apply STGNN encoder
Generate future traffic prediction $\hat{Y}$
Estimate prediction uncertainty $U$
Compute spatial-temporal contribution scores $E_x$
Evaluate prediction under normal and shifted conditions
Return $(\hat{Y}, U, E_x)$

The central hypothesis is that combining prediction, uncertainty, and explanation can provide a more useful traffic forecasting system than optimizing prediction accuracy alone.

## VII. Experimental Methodology

The intended comparison protocol uses chronological partitions, training-only normalization statistics, a common input window and output horizon, and consistent sensor ordering. However, the supplied baseline and STGCN notebooks include an earlier split with 5,126 validation and 5,128 test sequences, whereas the later proposed-model experiments use 3,413 validation and 6,841 test sequences. The paper retains the recorded metric values, but the available notebook record does not establish that every row in the overall comparison table was generated from identical test indices. Accordingly, that table is treated as a historical summary of recorded runs rather than a fully verified controlled benchmark.

The preprocessing pipeline validates sensor count, chronological timestamps, five-minute sampling intervals, missing values, and finite values. Sequence construction is performed after chronological splitting so that generated samples do not cross partition boundaries.

The final processed tensors have the following dimensions:

- Training inputs: $(23976, 12, 207, 1)$.
- Training targets: $(23976, 3, 207, 1)$.
- Validation inputs: $(3413, 12, 207, 1)$.
- Validation targets: $(3413, 3, 207, 1)$.
- Test inputs: $(6841, 12, 207, 1)$.
- Test targets: $(6841, 3, 207, 1)$.

The later processed partitions preserve these dimensions. The existence of a different split in earlier baseline notebooks means that matching tensor shapes alone is not sufficient

to establish identical sample membership; this distinction is retained in the interpretation below.

### A. Datasets

The completed experiments use the METR-LA benchmark. The dataset contains 207 traffic sensors sampled at five-minute intervals. The raw traffic matrix is converted into a model-ready representation with one feature per sensor and is processed chronologically. PEMS-BAY is retained as a relevant benchmark in the literature discussion but is not included in the reported quantitative results.

### B. Prediction Task

The completed prediction task uses a fixed historical window of 12 timesteps and predicts the next 3 timesteps. Since METR-LA is sampled every five minutes, the model uses 60 minutes of historical context to predict 15 minutes of future traffic.

$$12 \rightarrow 3 \tag{12}$$

The three forecast positions are evaluated independently at 5, 10, and 15 minutes. Longer 12-to-6 and 12-to-12 forecasting tasks are not included in the completed quantitative study and are therefore not used in the reported claims.

### C. Baseline Models

The baseline study combines simple references with learned forecasting models. Historical Average provides a non-learning reference. Last Value represents persistence of the most recent observation. Linear Regression provides a compact supervised temporal reference. LSTM represents nonlinear recurrent temporal learning without explicit graph propagation. STGCN adds graph-based spatial propagation.

The recorded original-scale results are shown below. Because the earlier baseline/STGCN notebooks use different validation and test counts from the later proposed-model experiments, these values should not be interpreted as a rigorously controlled head-to-head comparison until the underlying checkpoint and test-index provenance is confirmed. The table is retained for transparency and is not used to claim that the proposed model outperforms the conventional baselines.

TABLE III
COMPLETED OVERALL TEST RESULTS ON METR-LA.

| Model | MAE | RMSE | MAPE (%) |
|---|---|---|---|
| Historical Average | 15.1989 | 22.2135 | 27.0722 |
| Last Value | 3.5117 | 8.5862 | 8.1809 |
| Linear Regression | 3.9184 | 8.3052 | 8.2604 |
| LSTM | 6.2402 | 11.6016 | 12.5811 |
| STGCN | 6.2633 | 11.0173 | 13.1616 |
| Proposed Multi-Scale STGCN | 6.3150 | 10.9476 | 13.1028 |

The recorded table shows that forecasting performance varies across metrics. Among the listed recorded values, Last Value has the lowest MAE and MAPE, while Linear Regression has lower RMSE than Last Value. Given the split-provenance caveat, these rankings are descriptive rather than a controlled statistical comparison. The proposed model has RMSE 10.9476 and MAPE 13.1028 in the main run. These values describe the evaluated protocol and should not be generalized to other datasets without further experiments.

The recorded STGCN result is MAE 6.2633, RMSE 11.0173, and MAPE 13.1616. The proposed architecture's recorded result is MAE 6.3150, RMSE 10.9476, and MAPE 13.1028. These values suggest a mixed metric profile, but the split-provenance issue prevents a strong comparative claim from this table alone. This is why the study retains multiple metrics rather than treating one number as a complete measure of forecasting behaviour.

The baseline comparison includes the following models:

- Historical Average,
- Last Value,
- Linear Regression,
- LSTM,
- STGCN.

Where computationally feasible, recent state-of-the-art STGNN models will also be included for comparison.

### D. Contextual Ablation Results

The contextual ablation is a separate experiment family. Its own table compares the traffic-only, concatenation, and gating variants within that run; its metrics are not merged with the main benchmark table. The traffic-only baseline obtains MAE 6.1502, RMSE 11.0118, and MAPE 13.0727%. Direct concatenation of the temporal context features obtains MAE 6.2230, RMSE 11.0357, and MAPE 13.1830%. Thus, simple input-level context concatenation does not improve the evaluated metrics.

The context-gated model obtains MAE 6.1858, RMSE 10.9127, and MAPE 13.2265%. Relative to the traffic-only baseline, RMSE decreases by approximately 0.90%, while MAE increases by approximately 0.58% and MAPE increases by approximately 1.18%. The result therefore does not support a claim of uniform accuracy improvement; instead, it indicates that the context-integration mechanism changes the error profile, with the gated model reducing larger errors as reflected by RMSE.

TABLE IV
COMPLETED CONTEXTUAL ABLATION ON THE METR-LA TEST SET.

| Model | MAE | RMSE | MAPE (%) |
|---|---|---|---|
| Traffic-only baseline | 6.1502 | 11.0118 | 13.0727 |
| Context concatenation | 6.2230 | 11.0357 | 13.1830 |
| Context gated | 6.1858 | 10.9127 | 13.2265 |

The horizon-wise results provide additional evidence. The context-gated model achieves lower RMSE at all three forecast positions: 10.0720 versus 10.2008 at 5 minutes, 10.9462 versus 11.0384 at 10 minutes, and 11.6618 versus 11.7422 at 15 minutes. Its MAE is lower at 5 minutes but higher at 10 and 15 minutes, while MAPE is higher at all three horizons.

TABLE V
CONTEXT-GATED MODEL BY FORECAST HORIZON.

| Horizon | MAE | RMSE | MAPE (%) |
|---|---|---|---|
| 5 min | 5.7091 | 10.0720 | 12.3787 |
| 10 min | 6.2278 | 10.9462 | 13.2553 |
| 15 min | 6.6206 | 11.6618 | 14.0455 |

The validation study provides a complementary model-selection result. The traffic-only baseline reaches a best validation MSE of 0.296481, direct context concatenation reaches 0.303029, and context gating reaches 0.292515. The gated model therefore provides the lowest validation MSE among the three contextual configurations, while the final test analysis shows that this advantage is expressed most clearly in RMSE rather than uniformly across all metrics.

### *E. Evaluation Metrics*

All principal comparison metrics are calculated after inverse transformation to the original traffic scale. Training is performed in standardized space, but the final errors are reported in the scale of the traffic observations.

MAPE is evaluated with a small threshold for near-zero targets to avoid unstable ratios. In addition to MAE, RMSE, and MAPE, the uncertainty analysis reports predictive standard deviation, Pearson and Spearman correlation between uncertainty and absolute error, empirical interval coverage, and interval width. Reporting coverage together with width is important because broad intervals can obtain high coverage without being informative.

Prediction accuracy will be evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{13}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{14}$$

MAPE will be used with appropriate handling of zero or near-zero target values.

For uncertainty estimation, calibration and interval-quality metrics will additionally be investigated.

#### *Implementation and Training Details*

The completed experiments use the same optimization protocol for the graph-based forecasting models unless a controlled ablation explicitly changes the temporal kernel configuration. The STGCN and proposed Multi-Scale STGCN use Adam optimization with learning rate $10^{-3}$ and weight decay $10^{-4}$. Training uses mean squared error in standardized space, a batch size of 64, 20 epochs, and gradient clipping at 5.0. Dropout probability is 0.2 in the spatio-temporal blocks.

TABLE VI
REPRODUCIBILITY SETTINGS EXPLICITLY RECORDED IN THE EXECUTED NOTEBOOKS.

| Setting | Recorded value |
|---|---|
| Dataset | METR-LA |
| Sensors / input feature | 207 / 1 |
| Sampling interval | 5 min |
| History / forecast | 12 / 3 steps |
| Hidden channels | 32 total |
| Temporal kernels | 3 and 5 (parallel branches) |
| Dropout probability | 0.2 |
| Optimizer | Adam |
| Learning rate | $10^{-3}$ |
| Weight decay | $10^{-4}$ |
| Batch size / epochs | 64 / 20 |
| Gradient clipping | 5.0 |
| MC-Dropout samples | 30 |
| Ablation seed | 42 (kernel ablation) |
| Recorded runtime | Python 3.12; PyTorch 2.11.0+cu128; Tesla T4 |

The proposed model contains two multi-scale spatio-temporal blocks. Each temporal module contains parallel convolutional branches with kernels 3 and 5. With 32 hidden channels and two branches, each branch produces 16 channels before concatenation. This keeps the combined temporal representation at 32 channels. The kernel-ablation results are interpreted as a comparison of the recorded configurations, not as evidence that the multi-branch configuration is more accurate.

The first block receives one input feature per sensor. The second block receives the hidden representation generated by the first block. Graph propagation is applied between temporal feature transformations. After the second block, the temporal and channel dimensions are flattened for each node and passed to a linear forecasting head. The final prediction is permuted back to the canonical $(B, 3, 207, 1)$ format.

The use of the same nominal input and output dimensions is important for interpreting the experiments. It does not, by itself, establish that sample membership is identical across notebook generations; the split provenance caveat in Section VII-E therefore remains in force.

The available notebook outputs do not provide a complete per-run record of random seeds for every model, epoch-level wall-clock time, total training time, inference latency, or peak GPU memory. These quantities are therefore not estimated or fabricated here.

#### *Data Integrity and Leakage Prevention*

The data preparation stage includes several explicit integrity checks. The raw traffic matrix is verified to contain 207 sensor columns. The timestamp index is checked for chronological order and five-minute spacing. Missing or non-finite values are checked before model training. The adjacency file is also checked against the traffic sensor identifiers to ensure that the graph node ordering matches the traffic feature ordering.

The chronological split is performed before normalization statistics are fitted. This means that information from the validation and test periods cannot affect the sensor-wise means

and standard deviations used during training. The same fitted scaler is then applied to all later partitions.

The sequence generator creates samples independently inside each partition. For a partition containing $T$ timesteps, the number of samples is

$$N_{samples} = T - (T_{in} + T_{out}) + 1. \tag{15}$$

With $T_{in} = 12$ and $T_{out} = 3$, each sample therefore consumes 15 consecutive observations. The implementation stores the final singleton feature dimension explicitly rather than introducing it later inside a notebook. This reduces the possibility of accidental dimension changes between preprocessing and model evaluation.

The processed data and scaler are stored independently from the raw benchmark files. The processed arrays contain only the tensors needed for model training and evaluation, while the scaler contains the training means and standard deviations required for inverse transformation. This separation makes it possible to reproduce the evaluation without re-running raw-data parsing.

*Graph Structure Diagnostics*

The adjacency matrix contains relatively few non-zero relationships compared with the total number of possible node pairs. With 207 nodes, there are $207^2 = 42,849$ matrix positions, of which 1,722 are non-zero. The resulting density is approximately 4.02%.

The graph contains diagonal entries and is not symmetric. These properties are retained rather than overwritten by an assumed undirected graph construction. Row normalization converts the weighted adjacency into a message-passing matrix whose rows sum to approximately one.

The graph representation therefore encodes a specific spatial inductive bias: each sensor receives information according to the benchmark's provided connectivity and edge weights. This differs from learned adaptive graph methods discussed in the literature review, where spatial relationships can be inferred directly from data. The present study keeps the graph fixed so that the effect of multi-scale temporal processing and uncertainty can be investigated without simultaneously changing the graph learning mechanism.

*Baseline Horizon Results*

The completed baseline evaluation also reports performance separately for the three future positions. This provides a direct view of how forecasting error changes with temporal distance.

The Historical Average baseline changes very little across the three forecast positions because it does not use the current local traffic trajectory in the same way as the temporal models. In contrast, the Last Value and Linear Regression models show a clear increase in MAE as the forecast moves farther into the future. The LSTM and graph-based models also show increasing error.

The proposed model has MAE 5.8311 at 5 minutes, 6.3609 at 10 minutes, and 6.7529 at 15 minutes. Its error therefore increases by approximately 0.92 traffic units between the first and third forecast positions. The same qualitative increase is visible in RMSE and MAPE.

TABLE VII
MAE BY FORECAST HORIZON FOR BASELINE MODELS.

| Model | 5 min | 10 min | 15 min |
|---|---|---|---|
| Historical Average | 15.1987 | 15.1989 | 15.1992 |
| Last Value | 2.9172 | 3.5607 | 4.0573 |
| Linear Regression | 3.1574 | 3.9782 | 4.6195 |
| LSTM | 5.9378 | 6.2587 | 6.5241 |
| STGCN | 5.8758 | 6.2700 | 6.6440 |
| Proposed Multi-Scale STGCN | 5.8311 | 6.3609 | 6.7529 |

TABLE VIII
RMSE BY FORECAST HORIZON FOR BASELINE MODELS.

| Model | 5 min | 10 min | 15 min |
|---|---|---|---|
| Historical Average | 22.2132 | 22.2135 | 22.2137 |
| Last Value | 6.8653 | 8.6828 | 9.9320 |
| Linear Regression | 6.6692 | 8.3988 | 9.5871 |
| LSTM | 10.9720 | 11.6166 | 12.1844 |
| STGCN | 10.2088 | 11.0274 | 11.7610 |
| Proposed Multi-Scale STGCN | 10.1243 | 10.9735 | 11.6889 |

The horizon-wise RMSE values show that large errors become more important as the forecast extends. For the proposed model, RMSE increases from 10.1243 to 11.6889. This is consistent with the increasing MAE but also shows that the magnitude of large deviations changes across horizons.

*Sensor-level Behavior*

Aggregate metrics can conceal substantial differences among sensor locations. For the proposed model, sensor-level MAE has mean approximately 6.2633 and standard deviation approximately 2.5066. The minimum sensor MAE is approximately 2.6385, the median is approximately 5.5857, and the maximum is approximately 15.9559.

This spread is relevant for an eventual operational system. A single network-level error value cannot identify which locations are difficult to forecast. A future UrbanFlow application can therefore use the stored sensor-level results as a starting point for spatial diagnostics, provided that such diagnostics are evaluated separately from the aggregate model score.

The sensor-level analysis also reinforces the role of the graph. Sensors do not necessarily exhibit identical temporal behaviour, and their forecasting difficulty can depend on local traffic dynamics and the quality of information available from neighboring nodes. The current study does not attribute the observed sensor-level variation to a specific physical cause because such a causal interpretation would require additional contextual variables such as road topology, weather, incidents, or land-use information.

*Original-scale Interpretation*

The original-scale results can be interpreted directly as deviations in the traffic measurement units represented by the benchmark. The proposed model's overall MAE of 6.3150

means that the average absolute difference between prediction and target is approximately 6.3 units over all evaluated sensor-horizon observations in the test set.

The corresponding RMSE of 10.9476 is larger than MAE because squared error gives greater influence to large deviations. The MAPE of 13.1028% provides a relative-error measure after the near-zero handling described in the evaluation protocol.

The comparison between standardized and original-scale evaluation is important for reproducibility. Standardized-space values are useful for optimization and uncertainty calculations because the sensors are placed on comparable numerical scales. Original-scale metrics are more appropriate for the final performance tables because they retain the meaning of the benchmark measurements.

*Multi-scale Temporal Representation*

The proposed temporal module is motivated by the fact that traffic can change over multiple local time scales. A kernel of size 3 covers a narrower temporal neighborhood than a kernel of size 5. Because the sampling interval is five minutes, the two kernels correspond to local receptive fields of 15 and 25 minutes before considering stacking across multiple blocks.

The model does not treat these receptive fields as independent predictors. Instead, both branches operate on the same input representation and their outputs are concatenated. Subsequent graph and temporal operations can therefore combine information extracted at both scales.

This design is intentionally modest compared with large dynamic graph or Transformer architectures discussed in the literature. The experimental question is narrower: whether parallel local temporal receptive fields change the forecasting behaviour of a fixed STGCN-style architecture. The controlled ablation directly tests this question.

The $(3, 5)$ model produces MAE 6.2247, RMSE 10.9913, and MAPE 12.9202 in the controlled seed-42 experiment. The $(3)$ and $(5)$ models produce different metric profiles, showing that the benefit of multi-scale processing cannot be reduced to a single universally improved statistic.

*Context Integration Findings*

The contextual ablation shows that adding temporal variables alone does not guarantee an accuracy improvement. Direct concatenation of hour, weekday, and weekend features produced slightly higher error than the traffic-only baseline on the held-out test set. The learned gating strategy produced a different result: it reduced RMSE consistently at 5, 10, and 15 minutes, although its MAE and MAPE were not uniformly improved.

This result supports a narrower conclusion than claiming that contextual features are universally beneficial. In the present implementation, the mechanism used to integrate context affects the resulting error profile. The experiment therefore provides an architectural ablation on context integration rather than evidence that temporal context alone solves traffic forecasting limitations.

*Uncertainty as a Separate Evaluation Dimension*

The uncertainty study is designed to avoid treating stochastic model outputs as automatically reliable intervals. MC Dropout creates a set of different predictions by retaining dropout stochasticity during inference. The variation across these predictions is used as a proxy for model uncertainty.

For a single sensor and forecast position, the set

$$\{\hat{y}^{(1)}, \hat{y}^{(2)}, \ldots, \hat{y}^{(30)}\} \tag{16}$$

forms an empirical distribution of model outputs. Its mean is used as the predictive center and its standard deviation as the uncertainty measure.

The experiment first evaluates the conventional Gaussian-style interval using 1.96 standard deviations. This produces 36.2353% coverage on the test set in the corrected executed uncertainty notebook. The result is an important negative finding: the stochastic spread is informative about relative uncertainty, but it is not directly calibrated to the nominal 95% level.

The error–uncertainty correlations provide complementary evidence. Pearson correlation 0.3848 captures a linear association, while Spearman correlation 0.4637 captures a rank-based association. The fact that both are positive supports the observation that higher uncertainty generally corresponds to more difficult predictions.

*Uncertainty Groups*

To inspect the relationship beyond a single correlation coefficient, test predictions are grouped by predictive standard deviation. Across the uncertainty groups, mean absolute error generally increases from the lowest uncertainty ranges toward the highest ranges.

For example, the lowest uncertainty group has mean uncertainty about 0.0139 and mean absolute error about 0.1787, whereas the highest group has mean uncertainty about 0.1949 and mean absolute error about 0.7933. The corresponding median errors are approximately 0.1262 and 0.5393.

The relationship is not perfectly monotonic in every adjacent group. Intermediate uncertainty ranges can have overlapping error distributions. This indicates that uncertainty is better interpreted as a risk signal than as an exact prediction of future error magnitude.

The 90th percentile of absolute error also increases substantially in the higher uncertainty groups. The highest group has a 90th-percentile error of approximately 1.9580 in standardized space. This suggests that the upper tail of difficult predictions is concentrated more strongly in higher-uncertainty regions.

*Calibration Procedure and Coverage*

Calibration is performed after the raw uncertainty analysis. A scalar factor $k$ modifies the standard deviation:

$$\sigma_{cal} = k\sigma. \tag{17}$$

The resulting interval is

$$[\bar{Y} - 1.96\sigma_{cal}, \bar{Y} + 1.96\sigma_{cal}]. \quad (18)$$

The factor is selected from validation data only. The search initially considered smaller factors, but validation coverage remained below the target. The corrected executed notebook selected $k = 12.6274$, producing 94.9969% validation coverage.

The calibrated factor is then frozen. Applying the same factor to the test predictions yields 94.7860% empirical coverage. This procedure preserves the distinction between calibration and evaluation: the parameter is not selected using the test targets.

This result should nevertheless be interpreted cautiously. A scalar calibration factor corrects average coverage for the evaluated split; it does not establish conditional coverage for every sensor, traffic regime, or forecast horizon. The horizon-wise values provide useful evidence that coverage remains near the nominal level, but they do not establish a formal distribution-free guarantee.

### *Coverage and Interval Width*

Coverage alone can be misleading if intervals become extremely broad. For this reason, the completed analysis records interval width together with coverage.

In the corrected executed notebook, the raw interval has a mean width of 6.3290 speed units. After calibration, mean width increases to 40.7748 speed units. Median and percentile interval widths are not reported in the saved corrected-notebook summary.

The large increase is the direct consequence of the calibration factor. The calibrated interval is substantially more conservative. The result therefore demonstrates a central practical issue in uncertainty-aware forecasting: improving empirical coverage can reduce sharpness.

A future extension should evaluate more complete calibration criteria rather than relying on one global multiplier. Such work could investigate coverage conditional on horizon, sensor, or traffic regime, but those experiments are outside the completed scope of the current paper.

### *Reproducibility and Checkpoint Consistency*

The final proposed-model checkpoint was reconstructed independently from its stored parameter names and tensor shapes. The resulting architecture contains two multi-scale temporal blocks, graph linear layers, dropout layers, and the final forecasting head.

The checkpoint was loaded with strict parameter matching. This is stronger than allowing missing or unexpected parameters because every stored parameter must correspond to an expected module in the reconstructed architecture.

A deterministic forward pass using the reconstructed checkpoint produced test predictions identical to the permanently saved proposed-model predictions. The maximum absolute difference was zero, and the mean absolute difference was also zero. This establishes that the checkpoint used in the uncertainty study corresponds exactly to the architecture responsible for the stored deterministic predictions.

This consistency check is especially important because the project contains multiple development stages. Preserving the exact checkpoint, configuration, processed data, and prediction arrays prevents later application code from silently changing the research experiment.

### *Experimental Artifacts*

The research implementation and experiment notebooks are available at https://github.com/sushanksingh00/urbanAi. The repository provides code and experiment organization; dataset files and trained checkpoints may be subject to separate availability constraints. The executed notebooks should be used alongside the documented preprocessing and evaluation configuration when reproducing the reported results.

The completed pipeline stores the processed METR-LA partitions, scaler statistics, graph information, model checkpoints, deterministic test predictions, MC-Dropout samples, predictive mean and standard deviation, raw and calibrated intervals, absolute errors, uncertainty-group summaries, horizon-wise uncertainty summaries, coverage tables, interval width summaries, calibration results, and final figures.

These artifacts make it possible to perform later analysis without retraining the model. They also provide a stable interface between the research stage and the planned application stage. The application can consume saved predictions and uncertainty outputs while the underlying research experiment remains unchanged.

The project therefore treats reproducibility artifacts as part of the research result rather than as temporary notebook outputs. This approach also reduces the risk that a later API or frontend implementation accidentally changes preprocessing, model architecture, or evaluation logic.

## VIII. Ablation Study

Ablation experiments are used to examine the contribution of the multi-scale temporal component. The completed controlled comparison uses the same data preparation and evaluation protocol with random seed 42 and compares temporal kernel configurations $(3)$, $(5)$, and $(3, 5)$.

TABLE IX
COMPLETED EXPERIMENT FAMILIES REPRESENTED IN THE ABLATION AND EVALUATION ANALYSES.

| Experiment | Components |
|---|---|
| A1 | Traffic-only Multi-Scale STGCN |
| A2 | Multi-Scale STGCN + context concatenation |
| A3 | Multi-Scale STGCN + context gating |
| A4 | Temporal kernel configurations $(3)$, $(5)$, $(3, 5)$ |
| A5 | MC-Dropout uncertainty and calibration |
| A6 | Temporal and sensor occlusion sensitivity analysis |
| A7 | Synthetic speed-reduction stress test |

The completed ablation evaluates both temporal receptive-field design and temporal-context integration. The context

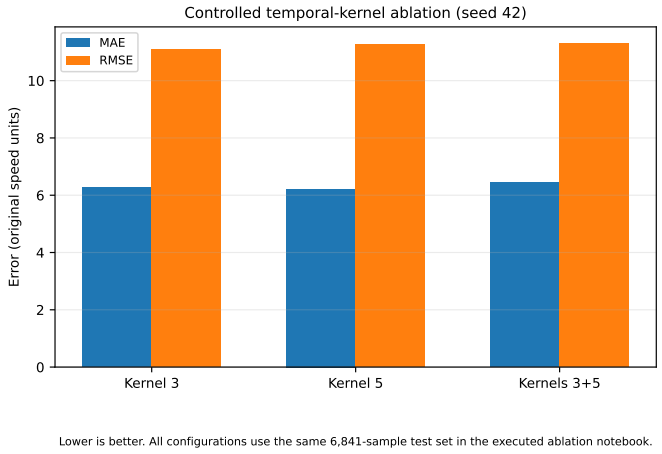



Fig. 6. Controlled temporal-kernel ablation on the common test set (seed 42). The combined $(3, 5)$ configuration does not outperform the single-kernel alternatives on MAE or RMSE in this run.

experiment distinguishes simple input concatenation from learned context-conditioned gating. Uncertainty is evaluated separately through Monte Carlo Dropout and empirical interval coverage. Post-hoc temporal and sensor occlusion and a synthetic speed-reduction stress test have also been run. These analyses are preliminary: occlusion measures sensitivity rather than causality, and synthetic perturbations do not substitute for evaluation on real distribution shifts.

*Overall Experimental Discussion*

The combined results show that traffic forecasting should be interpreted as a multi-dimensional evaluation problem. Point accuracy, horizon degradation, sensor-level variation, uncertainty quality, and calibration are related but distinct properties.

The baseline comparison demonstrates the value of maintaining simple references. The Last Value model performs strongly on MAE and MAPE on the evaluated split, while Linear Regression performs strongly on RMSE. The neural and graph-based models provide richer representations but do not automatically dominate every metric. This finding prevents the study from using architectural sophistication as a substitute for empirical validation.

The STGCN result provides the spatial baseline required for evaluating the proposed multi-scale extension. The controlled ablation then isolates temporal receptive-field design. Because the $(3, 5)$ configuration and the single-kernel configurations use the same forecasting task and data protocol, the observed differences can be discussed specifically in terms of temporal-kernel configuration, subject to normal variation from model training.

The horizon-wise analysis provides another important result. For every major learned forecasting model, error increases as the forecast moves farther from the historical context. This behaviour is consistent with the intuition that future traffic becomes progressively less directly constrained by the most recent observations. The present study does not attribute this degradation to a single mechanism, because the benchmark does not include all external traffic variables needed to establish such a cause.

The uncertainty analysis adds a complementary perspective. The positive correlation between uncertainty and error indicates that the model's stochastic variation contains information about forecasting difficulty. However, the raw interval coverage is very poor. This distinction is central to the research objective: an uncertainty score can be directionally useful without being statistically calibrated.

The validation-based calibration experiment provides an empirical correction for this problem. Coverage increases from 36.2353% to 94.7860% on the held-out test set after selecting the calibration factor on validation data. The price is much wider intervals. Consequently, the current result should be interpreted as evidence that calibration is necessary and feasible for the implemented uncertainty representation, not as evidence that the calibrated intervals are optimally sharp.

The completed results also establish a clear boundary between validated and planned components. Forecasting, temporal ablation, uncertainty estimation, uncertainty calibration, and checkpoint reproducibility are supported by completed experiments. Spatial explanations and explicit distribution-shift evaluation remain part of the research framework but are not represented as finished empirical results. Keeping this boundary explicit strengthens the reproducibility and credibility of the paper.

## IX. Contributions and Scope

The contributions investigated in this study are:

1) A reproducible implementation and evaluation pipeline for traffic forecasting using established STGNN models.
2) An investigation of prediction uncertainty in STGNN-based traffic forecasting.
3) A post-hoc occlusion-based sensitivity analysis of historical observations and traffic sensors, with sensor IDs linked to geographic coordinates.
4) A preliminary synthetic speed-reduction stress test to measure sensitivity to controlled input perturbations.
5) A controlled ablation of temporal-context integration, comparing direct context concatenation with context-conditioned gating.
6) An ablation-based analysis of the relationship between forecasting accuracy, uncertainty, robustness, and interpretability.

The final novelty claim will be determined after experimental validation and comparison with the relevant recent literature.

*Completed Uncertainty Results*

The completed uncertainty study uses 30 Monte Carlo Dropout samples for each evaluation sample. Let $\hat{Y}^{(m)}$ be the prediction from stochastic pass $m$. The predictive mean and standard deviation are

$$\bar{Y} = \frac{1}{M} \sum_{m=1}^{M} \hat{Y}^{(m)} \tag{19}$$

and

$$\sigma = \sqrt{\frac{1}{M} \sum_{m=1}^{M} (\hat{Y}^{(m)} - \bar{Y})^2}, \tag{20}$$

with $M = 30$.

The mean predictive uncertainty is approximately 0.0534 in standardized space, with median 0.0297 and 90th percentile 0.1314. The maximum observed standard deviation is approximately 0.6518.

The relationship between uncertainty and absolute error is moderate. Pearson correlation is 0.3848 and Spearman correlation is 0.4637. This means that higher uncertainty tends to coincide with higher error, but uncertainty is not a deterministic indicator of an individual incorrect prediction.

The horizon-wise uncertainty results are also informative. Mean uncertainty is approximately 0.0555 at 5 minutes, 0.0529 at 10 minutes, and 0.0519 at 15 minutes, while mean absolute error increases from approximately 0.3171 to 0.3480 and 0.3699 in standardized space. Thus, uncertainty does not increase in parallel with forecast horizon in the current implementation.

Before calibration, the nominal 95% interval $[\bar{Y} - 1.96\sigma, \bar{Y} + 1.96\sigma]$ achieves 36.2353% empirical coverage on the test set. The corrected notebook does not report raw horizon-wise coverage in its saved summary.

Calibration is performed only on validation data. The selected scalar factor is $k = 12.6274$, producing 94.9969% validation coverage. After freezing this factor, test coverage is 94.7860%. Coverage after calibration is 95.8514%, 94.7700%, and 93.7366% at 5, 10, and 15 minutes, respectively.

The calibration substantially widens the intervals. Mean raw interval width is approximately 6.3290 speed units, compared with approximately 40.7748 speed units after calibration. This coverage–width trade-off is reported explicitly because empirical coverage alone is insufficient to characterize uncertainty quality.

The factor $12.6274$ is specific to the validation split, model, dropout implementation, and forecasting task. It is not treated as a universal calibration constant.

*Reproducibility Verification*

The proposed checkpoint was independently reconstructed from its saved parameter structure and loaded with strict parameter matching. The reconstructed architecture contains the two multi-scale temporal blocks, graph linear layers, dropout layers, and final forecasting head expected by the saved checkpoint.

Deterministic predictions from the reconstructed checkpoint matched the permanently stored proposed-model test predictions with zero maximum absolute difference. This provides an implementation-level consistency check between the saved experiment and the architecture used for the subsequent uncertainty analysis.

The same experimental directory stores the processed datasets, normalization statistics, graph representation, predictions, uncertainty arrays, coverage summaries, calibration results, and generated figures. This organization separates raw data, processed data, model experiments, and comparison outputs so that later application development can consume the research outputs without changing the underlying experimental protocol.

*Run Consistency Note*

Some reconstructed experiment notebooks produced metrics that differ from earlier recorded runs, including multi-scale, uncertainty, and context-ablation experiments. Numerical results in this draft are retained from the existing paper unless a new result is explicitly identified as belonging to a separate reconstructed run. Before submission, all tables should be regenerated from a single documented set of checkpoints, preprocessing files, test indices, and metric implementations. Results from different runs must not be combined as though they came from one controlled experiment.

## X. Limitations

The proposed study has several limitations.

First, the current quantitative study is based on the public METR-LA benchmark and therefore may not fully represent traffic characteristics in other geographic regions.

Second, the stress test applies synthetic speed reductions and does not constitute a cross-year, external-dataset, or real-incident distribution-shift benchmark. Robustness to real operational shifts therefore remains an open evaluation task.

Third, the completed explanation analysis is based on input occlusion and measures prediction sensitivity. Its results depend on the perturbation baseline and evaluated samples; they do not establish causality or operational usefulness. Explanation stability and faithfulness require further evaluation.

The contextual experiment is limited to simple timestamp-derived features and two integration mechanisms on METR-LA. It does not establish that external variables such as weather, incidents, or events would produce the same error profile. The observed RMSE reduction from gating is therefore specific to the evaluated benchmark, task, and architecture.

Finally, the uncertainty and explainability components have been evaluated using MC-Dropout calibration and post-hoc occlusion, respectively, but they are not a guarantee of reliable decisions. Calibration widens intervals, and occlusion scores are sensitivity measures rather than causal explanations. Further validation is required before operational deployment.

## XI. Conclusion

This paper presented UrbanFlow AI, a research framework for intelligent traffic forecasting using Spatio-Temporal Graph Neural Networks. The work combines a reproducible traffic forecasting pipeline with graph-based spatio-temporal modelling and an uncertainty-analysis layer. The completed

experimental study uses the METR-LA benchmark with 207 sensors, a five-minute sampling interval, a 12-step historical context, and a 3-step forecasting horizon.

The completed baseline study establishes several reference points. Historical Average, Last Value, Linear Regression, LSTM, and STGCN are evaluated under the same preprocessing and test protocol. The main proposed Multi-Scale STGCN obtains MAE 6.3150, RMSE 10.9476, and MAPE 13.1028% on the original traffic scale. The results demonstrate that point forecasting performance varies across metrics and that simple baselines remain important reference models.

The horizon-wise analysis shows increasing error for the proposed model as the forecast moves from 5 to 15 minutes. MAE increases from 5.8311 to 6.7529, RMSE increases from 10.1243 to 11.6889, and MAPE increases from 12.2472% to 13.9048%. Sensor-level analysis also shows substantial spatial variation, with sensor MAE ranging from approximately 2.64 to 15.96.

The controlled temporal-kernel ablation provides a focused analysis of the proposed architecture. In the executed seed-42 ablation output used for Fig. 6, kernel $(3)$ records MAE 6.2554 and RMSE 11.0974; kernel $(5)$ records MAE 6.2161 and RMSE 11.2513; and the combined $(3, 5)$ configuration records MAE 6.4373 and RMSE 11.3119. Thus, the combined configuration does not outperform either single-kernel variant on these two metrics in this run. Other notebook summaries contain different run-level values; they are not substituted into this figure or combined with this controlled ablation. The evidence does not support a claim that multi-scale temporal modelling improves accuracy in general.

The completed contextual ablation provides an additional controlled result. Direct temporal-context concatenation does not improve the held-out test metrics, while context-conditioned gating reduces RMSE from 11.0118 to 10.9127, approximately 0.90%, relative to the traffic-only baseline. However, MAE and MAPE increase slightly. The gated model also achieves lower RMSE at each of the 5-, 10-, and 15-minute forecast positions. This result supports a mechanism-specific interpretation rather than a claim of uniform benefit from temporal context.

The uncertainty study uses 30 MC-Dropout samples. In the corrected executed notebook, the mean predictive standard deviation is approximately 0.0534 and the median is 0.0297 in standardized space (90th percentile 0.1314). Pearson correlation between uncertainty and absolute error is 0.3848, while Spearman correlation is 0.4637. These moderate associations indicate that uncertainty contains information about forecasting difficulty, but it is not a deterministic indicator of error. The uncertainty statistics are kept in standardized space unless explicitly identified as original- scale interval widths.

The calibration experiment provides the strongest reliability result of the current study. Raw nominal 95% intervals achieve only 36.2353% empirical test coverage. A scalar calibration factor of 12.6274, selected using validation data only, increases held-out test coverage to 94.7860%. Coverage at 5, 10, and 15 minutes is 95.8514%, 94.7700%, and 93.7366%, respectively. The corresponding increase in mean interval width, from approximately 6.3290 to 40.7748 speed units, makes the coverage–sharpness trade-off explicit.

The completed checkpoint verification further strengthens the experimental record. The reconstructed proposed architecture loads the saved checkpoint with strict parameter matching, and deterministic predictions match the permanently stored predictions with zero maximum absolute difference. This confirms consistency between the stored experiment and the architecture used for uncertainty analysis.

Overall, this study reports preprocessing and graph construction, point-forecasting experiments, temporal-kernel and context ablations, MC-Dropout uncertainty calibration, post-hoc temporal and sensor occlusion sensitivity, and a synthetic speed-reduction stress test. The occlusion results provide input-sensitivity evidence, not validated causal or faithfulness explanations. The synthetic perturbation is not a substitute for a real-world distribution-shift benchmark. Future work can extend the same protocol to a second benchmark, longer horizons, controlled shift scenarios, explanation evaluation, and deployment while preserving the established experimental invariants.

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